

5 (a) State what is meant by a meson.

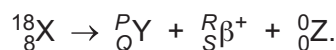
.....
..... [1]

(b) The elementary charge is e .

Explain, in terms of quark composition, how a baryon may have an overall charge of:

- $-1e$
.....
- $+2e$
..... [3]

(c) The nuclide $^{18}_8\text{X}$ undergoes β^+ decay to produce nuclide ^P_QY and one other particle Z, as shown in the equation



(i) State the values of P , Q , R and S .

$P =$
 $Q =$
 $R =$
 $S =$ [2]

(ii) State the name of particle Z.

..... [1]

(iii) State the net change in quark composition as nuclide X turns into nuclide Y.

..... [1]

- (iv) Explain why the emission of particle Z results in the β^+ particles from the decays of different nuclei of $^{18}_8\text{X}$ having a continuous range of kinetic energies.

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..... [2]

[Total: 10]